DIC14
Schmitt Triggers

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 14 (PDF pages 58–61) to compare a CMOS inverter with the six-transistor Schmitt trigger, examine feedback sizing versus hysteresis/noise/delay, and test a distinct topology.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: Fig.14.1, three feedback sizes, bidirectional DC (58–59,61)	01_hysteresis, six cases	Independent sweeps avoid coupled branch jumps
Step 2: noise immunity (59–61)	02_noise_immunity	Same finite-duration disturbances
Step 3: speed comparison (60–61)	03_delay_comparison	Equal 8 fF and input edges
Question: alternate topology (61)	ALT; 05_alternative_dc	DC, noise and delay all tested
Additional numerical check	04_slow_ramp_check	1/10 μs ramps versus DC

3 Circuit and design derivation

The main series PMOS devices are each 3.6/0.25 μm and NMOS each 1.2/0.25 μm. Feedback PMOS/NMOS widths are 1.8/0.6, 3.6/1.2 and 5.4/1.8 μm for the three cases, all at L=0.25. The Fig.14.1 feedback connections are retained. The reference inverter uses widths 1.8/0.6 μm. The alternative is a strong input inverter ($W_p/W_n = 3.6/1.2$) driving two weak cross-coupled inverters (each 1.8/0.6). It also uses six MOS devices but a different feedback topology.

V^+ is the rising-input point where output falls through half supply; V^- is the falling-input point. Hysteresis width is $H = V^+ - V^-$. This state-dependent two-branch behavior must not be equated

with the two unity-gain margins of a single-valued inverter. The handout equations are square-law approximations; internal V_T and mobility are not independently verified here, so they are not treated as exact design equations.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

DC sweeps 0–1 V and 1–0 V in 0.5 mV steps. Multiple .dc statements in the source did not reliably replace the intended reverse sweep; explicit start/stop/step parameters now define both directions. Parallel hysteretic instances also exhibited coupled Newton branch jumps, so DC uses one feedback case at a time. Bidirectional 1 and 10 μs/V transient ramps provide a cross-check. Noise stimulus matches Lab8; delay comparison uses 0.1 ns edges, 8 fF output loads and 50% crossings.

5 Results and specification checks

All five decks and 12 cases completed. Thresholds below are in V; ramp values use 10 μs/V.

Circuit	V_{DC}^{+}	V_{DC}^{-}	H_{DC}	V_{ramp}^{+}	V_{ramp}^{-}
SCH 1	0.5922	0.3968	0.1955	0.5928	0.3961
SCH 2	0.6332	0.3573	0.2760	0.6341	0.3564
SCH 3	0.6552	0.3353	0.3200	0.6565	0.3338
Alt	0.6321	0.0879	0.5441	0.6325	0.0873

Circuit	t_r (ps)	t_f (ps)	t_{pHL} (ps)	t_{pLH} (ps)
INV	106.95	51.92	63.86	88.77
SCH 1	219.10	84.35	149.36	203.59
SCH 2	260.52	101.05	196.67	230.75
SCH 3	301.50	118.33	244.97	256.13
Alt	427.75	106.61	70.70	311.56

6 Discussion

Stronger feedback increases DC hysteresis from about 0.195 to 0.320 V while increasing delay. The slower-ramp thresholds differ from independent DC by at most about 1.5 mV, a finite-speed and numerical check rather than proof of the zero-rate limit. A 0.6 V, 0.9 ns disturbance can switch the inverter while the Schmitt retains state during that interval. A short pulse near the DC threshold

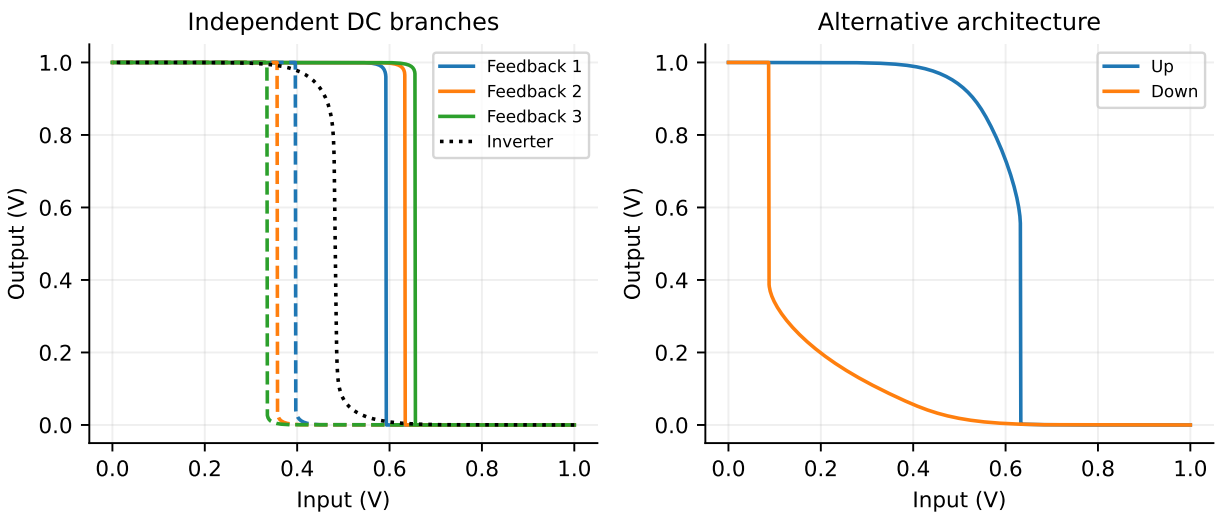


Figure 1: Independent DC hysteresis for three feedback strengths and the alternative; redrawn from this run's HSPICE data.

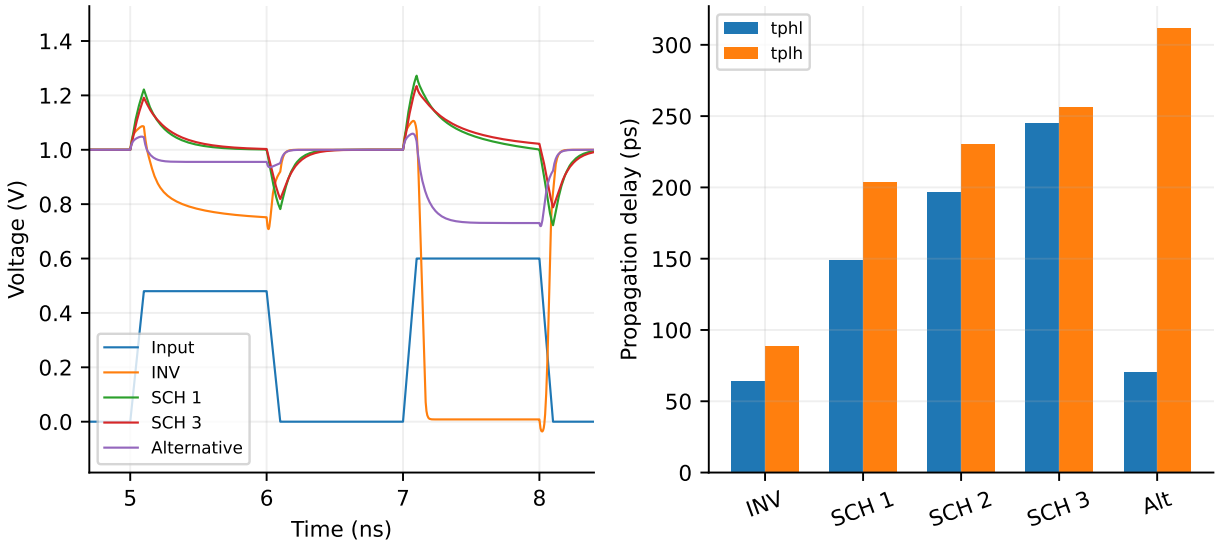


Figure 2: Finite-duration noise and propagation delays with equal loads; redrawn from this run's HSPICE data.

may not allow regeneration to finish; this does not establish a DC margin beyond the measured threshold. Actual data include temporary rail excursions whose reliability is unverified.

The alternative has about 0.544 V hysteresis but a falling-input threshold near 0.088 V, giving asymmetric noise behavior and a slower rising output. Larger hysteresis is therefore not universally better. Feedback strength must be evaluated together with speed, recovery conditions and input loading.

7 Conclusion

All three steps and the alternative-topology question have DC, transient and numerical cross-checks. The speed cost of stronger feedback and asymmetry of the alternative are explicit. Original-process voltage values and unverified reliability limits are not declared satisfied.

8 MOS transistor schematics

Source: SCH in `01_hysteresis.sp`.

The ALT comparison is from `05_alternative_dc.sp`. Both topologies are shown because their feedback connections differ. Here vp is V_{DD} and vn is ground.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

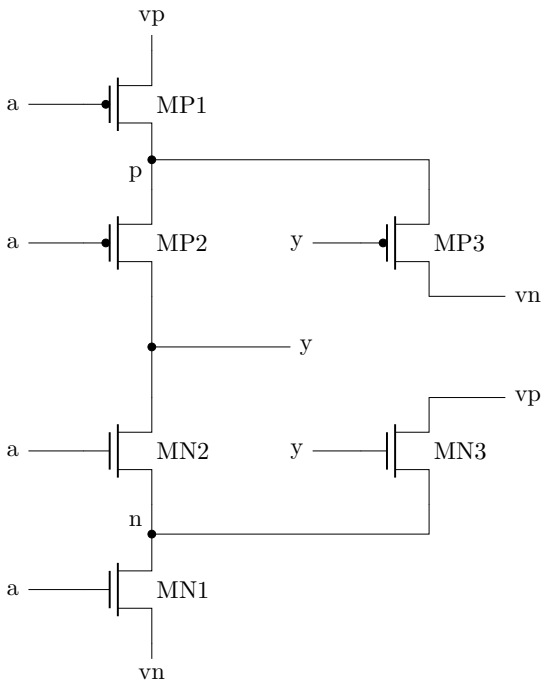


Figure 3: Complete six-MOS SCH. MP1/MP2: $W = 3.6\,\mu$m; MN1/MN2: $W = 1.2\,\mu$m; MP3: $W = 1.8f\,\mu$m; MN3: $W = 0.6f\,\mu$m. All $L = 0.25\,\mu$m. Feedback source/drain assignments retain the netlist: MP3 $(D, G, S, B) = (vn, y, p, vp)$; MN3 $(D, G, S, B) = (vp, y, n, vn)$.

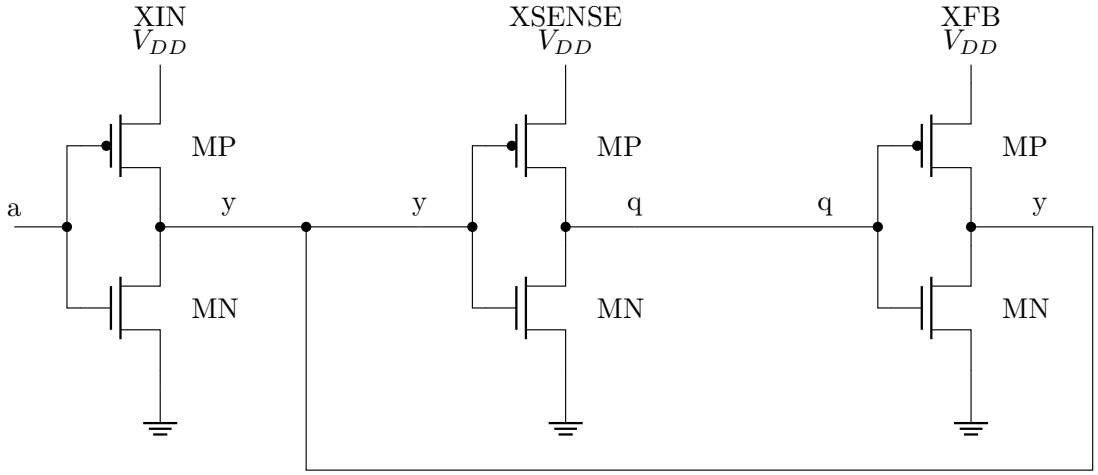


Figure 4: ALT: strong input inverter XIN ($W_n/W_p = 1.2/3.6\,\mu$m) and two weaker cross-coupled inverters XSENSE/XFB ($0.6/1.8\,\mu$m). The XFB output connects back to y; all lengths are $0.25\,\mu$m.